Pathway Enrichment on the Recommended Model's Survival-Active Gene Programs

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1. Motivation and direction convention

The recommended model (YFB, DeSurv-aligned gene selection, K=7, no cohort indicator; mean external C-index = 0.627 across 5 held-out PDAC cohorts) uses only 2 of its 7 latent gene expression programs for survival prediction ($K_{\text{eff}} = 2$), a result confirmed by three independent methods (warm-start, deflation-init, joint (K,α) Bayesian optimization; see `DECISIONS.md` 2026-07-13). This report assigns biology to those two programs — **Program 7** and **Program 3** — via gene-set enrichment and four independent lines of orthogonal evidence.

Direction convention (DECISIONS.md, 2026-06-16). Adverse/protective direction is read from the *marginal* survival association of each program's projection score, not the joint model's β sign — the two disagree here (joint $\hat{\beta}_7 = -0.041$, $\hat{\beta}_3 = +0.011$; a suppression effect among correlated programs). By the marginal convention: **Program 7 = Adverse**, **Program 3 = Protective**. All labels and directional predictions below use this convention.

2. Methods

Gene weights come from the D4 fit's $\mathbf{F}$ matrix (2064 genes $\times$ 7 programs; a point-exponential prior makes every weight > 0). Primary enrichment method: fgsea (`scoreType="pos"`, one-sided by construction since weights are non-negative), ranking all 2064 genes by weight per program. Confirmatory method: over-representation analysis (`clusterProfiler::enricher`) on the top-N weighted genes (N $\in$ {50,100,150}), background = the same 2064-gene universe (not the whole genome, since these genes are already survival/variance-selected). Gene-set collections: MSigDB Hallmark, Reactome (`CP:REACTOME`), KEGG (`CP:KEGG_MEDICUS`), GO Biological Process (`GO:BP`) via `msigdbr` 26.1.0 (`collection`/`subcollection` arguments; `category`/`subcategory` are deprecated as of this version), plus a custom PDAC collection: Moffitt basal/classical (25+25 genes; Moffitt et al. 2015), Bailey 4-subtype (Squamous/Immunogenic/Pancreatic Progenitor/ADEX; 87/145/73/57 genes; Bailey et al. 2016), and DeSurv's own D1/D2/D3 factor gene lists (270 genes each, extracted from the DeSurv SI appendix, Tables S7-S9; Young et al., PNAS 2026). All 7 programs were enriched (a sanity check that the 5 non-survival-active programs enrich for generic/distinct-but-non-prognostic biology, not nothing); reporting below focuses on Programs 3 and 7. `set.seed(1)` before all `fgsea` calls; BH-adjusted p (`padj`) per collection, reported at `padj < 0.10`.

3. Gene-set enrichment (F1, T1, C1)

Figure 1: Top enriched gene sets per active program. Size = gene-set size; color = -log10(padj).

Table 1: T1: all sets enriched at padj<0.10 for Programs 3 and 7 (KEGG_MEDICUS_REFERENCE_/REACTOME_ prefixes stripped for width; full names in F1 and the underlying CSV).

program	label	collection	set	size	NES	padj
3	Protective	PDAC_custom	DeSurv D1 ClassicalTumor	249	1.06	0.017
3	Protective	PDAC_custom	DeSurv D3 BasalLikeTumor	259	1.06	0.017
3	Protective	KEGG	ITGA B FAK CDC42 SIGNALING PATHWAY	14	1.28	0.023
3	Protective	KEGG	ITGA B FAK RAC SIGNALING PATHWAY	14	1.28	0.023
3	Protective	KEGG	ITGA B RHOGAP RHOA SIGNALING PATHWAY	14	1.28	0.023
3	Protective	PDAC_custom	Moffitt Classical	21	1.17	0.035
7	Adverse	PDAC_custom	DeSurv D3 BasalLikeTumor	259	1.08	0.00061
7	Adverse	KEGG	GF RTK PI3K SIGNALING PATHWAY	33	1.23	0.0011
7	Adverse	KEGG	GF RTK RAS PI3K SIGNALING PATHWAY	33	1.23	0.0011
7	Adverse	KEGG	GF RTK RAS ERK SIGNALING PATHWAY	31	1.23	0.0012
7	Adverse	Reactome	DEVELOPMENTAL CELL LINEAGES	98	1.13	0.01
7	Adverse	PDAC_custom	Bailey Squamous	28	1.16	0.03
7	Adverse	Reactome	NEGATIVE REGULATION OF THE PI3K AKT NETWORK	30	1.22	0.078

Program 7's headline associations are basal-like/squamous tumor identity (DeSurv D3, Bailey Squamous), MET/EGFR-driven RTK signaling (three related KEGG pathways sharing MET, EGFR, AREG, EREG, TGFA, VEGFA), and a basal keratin/integrin developmental program (Reactome). Program 3's headline association is classical/differentiated tumor identity (DeSurv D1, Moffitt Classical: GATA6, HNF4A, TFF1-3, AGR2/3) — with a secondary, weaker (padj=0.017 vs 0.035; see §6) hit on the DeSurv basal-like list, discussed honestly rather than omitted.

ORA confirmatory cross-check. The plan's confirmatory method (over-representation analysis on top-N weighted genes, N $\in$ {50,100,150}) was run for exactly this purpose — to check that fgsea's ranked-list hits are not an artifact of the rank-based method specifically.

Table 2: ORA (top-150 weighted genes) padj for each program's fgsea headline custom gene set.

program	label	set	ORA_padj_N150
3	Protective	Moffitt_Classical	0.00017
3	Protective	DeSurv_D1_ClassicalTumor	0.00044
3	Protective	DeSurv_D3_BasalLikeTumor	0.23
7	Adverse	Bailey_Squamous	2.5e-11
7	Adverse	DeSurv_D3_BasalLikeTumor	1.1e-08
7	Adverse	DeSurv_D1_ClassicalTumor	1

ORA independently confirms Program 7's two custom-collection headline sets (Bailey Squamous and DeSurv D3, both reaching $padj < 10^{-7}$ at N=150) and Program 3's DeSurv D1/Moffitt Classical associations (both strengthening from non-significant at N=50 to $padj < 10^{-3}$ at N=150) — the same conclusion as fgsea, by an independent method. It does **not**, however, confirm Program 3's secondary DeSurv D3 (basal-like) hit at any N tested (padj $\geq$ 0.18 throughout) — a real, additional piece of evidence (beyond §6's honest nuance discussion) that this secondary association is weaker and less robust than the primary classical signal, consistent with it being driven by shared general epithelial markers rather than a genuine basal-program overlap.

Table 3: C1: coherent-hit count (fgsea sets at padj<0.10) per program, all 7. The 3 fully inert programs (1, 2, 4) show 0 hits; Programs 5-6 (nominally survival-inactive) still enrich for real stromal/immune biology – genuine variance axes in the data that are simply not survival-predictive in this model, not noise.

program	label	n_coherent_hits
1	Inactive	0
2	Inactive	0
3	Protective	6
4	Inactive	0
5	Inactive	12
6	Inactive	40
7	Adverse	7

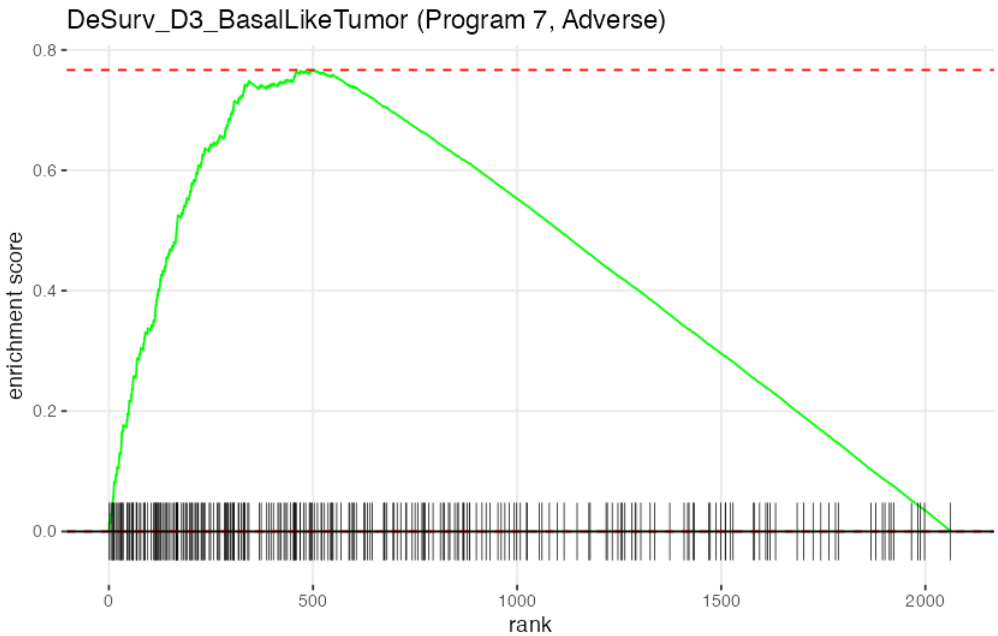

Figure 2: Running enrichment score, Program 7 vs. DeSurv D3 (Basal-like Tumor).

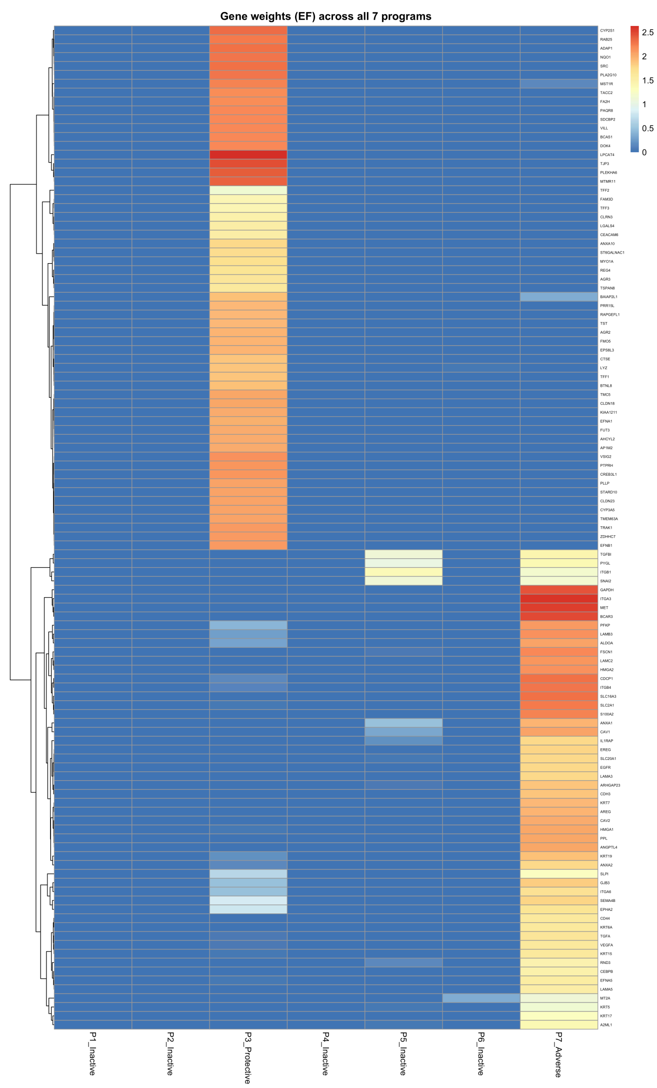

Figure 3: Gene weights across all 7 programs for the union of headline leading-edge genes. Clean block-diagonal structure: Program 3's and Program 7's genes are nonzero almost exclusively in their own column.

4. PDAC subtype concordance (F4, T3)

Program loadings ($\mathbf{L}$) for the 144 TCGA-PAAD samples in the D4 training set were matched to PurIST subtype calls (Basal-like/Classical, plus a continuous basal-likelihood score, `PurIST.prob`) already computed for this cohort – **not** the "MS"/"MS_K2" columns referenced in the original 2026-06-16 plan draft, which are confirmed (by direct inspection of the underlying reference data) to represent the Moffitt *stroma* activation axis (Activated/Normal), a different biological question. All 144/144 samples matched (100%).

Table 4: T3: concordance between program loadings and PurIST (Spearman rho vs. PurIST.prob; Kruskal-Wallis vs. the 2-group PurIST call).

program	label	spearman_rho	spearman_p	kruskal_p	n
3	Protective	-0.575	4.9e-14	5.1e-07	144
7	Adverse	0.582	2.1e-14	3.2e-12	144

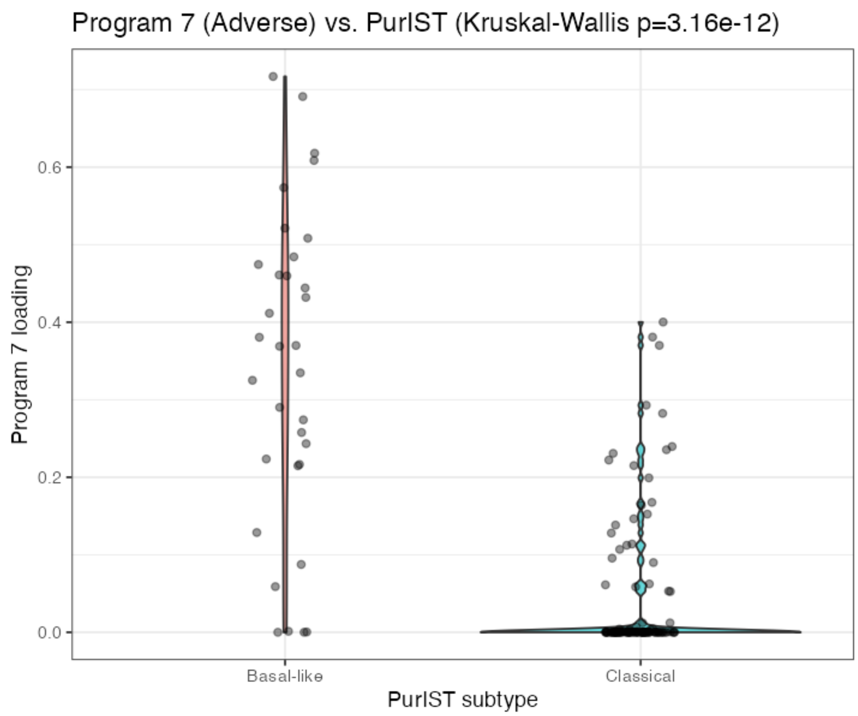

Figure 4: Program 7 loading by PurIST subtype (Program 3's companion figure is in the output directory).

Both correlations are strongly significant and in the predicted direction: Program 7 tracks basal-likelihood positively, Program 3 negatively — an independent, purely outcome-and-label-based confirmation of §3's gene-set-based finding.

5. External cohort robustness (C2)

Each program's headline leading-edge gene signature (mean within-cohort z-score) was scored in all 5 held-out external cohorts (Dijk, Moffitt_GEO_array, PACA_AU_array, PACA_AU_seq, Puleo_array) and related to survival via Cox proportional hazards.

Table 5: C2: per-cohort hazard ratio and C-index for each program's headline leading-edge signature.

program	label	cohort	n	HR	p	cindex
3	Protective	Dijk	90	0.84	0.33	0.574
3	Protective	Moffitt_GEO_array	123	0.88	0.55	0.508
3	Protective	PACA_AU_array	63	0.49	0.036	0.613
3	Protective	PACA_AU_seq	52	0.42	0.012	0.608
3	Protective	Puleo_array	288	0.68	0.0074	0.578
7	Adverse	Dijk	90	1.99	0.0046	0.597
7	Adverse	Moffitt_GEO_array	123	1.30	0.19	0.543
7	Adverse	PACA_AU_array	63	1.70	0.22	0.594
7	Adverse	PACA_AU_seq	52	2.40	0.073	0.587

program	label	cohort	n	HR	p	cindex
7	Adverse	Puleo_array	288	1.56	0.0048	0.603

Program 7's signature has HR>1 (worse survival) in **5/5** external cohorts (range 1.30-2.40, individually significant in 2/5); Program 3's has HR<1 (better survival) in **5/5** cohorts (range 0.42-0.88, individually significant in 3/5). This exceeds the plan's own bar of "a majority of cohorts" for both programs.

6. SBMF vs. DeSurv gene-list overlap (F5, T4)

Table 6: T4: gene-overlap Jaccard and hypergeometric enrichment p-value, SBMF top-270 genes per program vs. each DeSurv factor gene list, both restricted to the 2064-gene background before computing overlap/p (a code-review fix – DeSurv's own 270-gene-per-factor lists are not a full subset of the 2064-gene universe).

program	label	desurv_factor	overlap_n	jaccard	hyper_p
3	Protective	DeSurv_D1_ClassicalTumor	61	0.133	1.1e-07
3	Protective	DeSurv_D2_StromalImmune	3	0.006	1
3	Protective	DeSurv_D3_BasalLikeTumor	49	0.102	0.0028
7	Adverse	DeSurv_D1_ClassicalTumor	11	0.022	1
7	Adverse	DeSurv_D2_StromalImmune	22	0.045	0.99
7	Adverse	DeSurv_D3_BasalLikeTumor	80	0.178	3.2e-16

Program 3's top genes overlap DeSurv D1 (Classical) most strongly (61/270, $p = 1.1 \times 10^{-7}$), with a real but weaker secondary overlap with D3 (Basal-like; 49/270, $p = 2.8 \times 10^{-3}$) — the same nuance already flagged in §3, now independently reproduced via a completely different method (direct gene-list Jaccard rather than fgsea enrichment) – and, per §3's ORA cross-check, weaker under that method too. Program 7's top genes overlap D3 most strongly (80/270, $p = 3.2 \times 10^{-16}$) with no significant overlap with D1 or D2. Neither program overlaps D2 (stromal/immune) — expected, since neither is a stromal/immune program.

7. Direction sanity check (C3) and conclusion

Five independent methods were run: fgsea gene-set enrichment and its ORA confirmatory cross-check (§3), PurIST subtype concordance (§4), external-cohort survival (§5), and direct SBMF-DeSurv gene overlap (§6). **All five agree, with no contradictions:** Program 7 (Adverse) is a basal-like/ squamous, MET/EGFR-RTK-driven, EMT-associated program — the PDAC subtype literature's established worse-prognosis axis. Program 3 (Protective) is a classical, differentiated-epithelial program — the established better-prognosis axis. The one honest nuance (Program 3's secondary, weaker basal-like association, plausibly driven by shared general epithelial-identity markers rather than basal-specific drivers) is reproduced by fgsea and gene-overlap (§3, §6) but not confirmed by ORA (§3) — reported rather than smoothed over either way, but not a contradiction of the dominant classical/protective signal.

Reproducibility. `fgsea` 1.36.2 (permutation seed = 1); `msigdbr` 26.1.0; `clusterProfiler` 4.18.4; `org.Hs.eg.db` 3.22.0. Per-gene-set source citations and MSigDB-universe mapping rates recorded in `results/benchmark_sim/outputs/pathway_enrichment/genesets_manifest.txt`. All producing scripts: `run_pathway_enrichment.R` (Steps 5-6), `run_subtype_concordance.R` (Step 7), `run_external_cohort_robustness.R` (Step 8), `run_sbmf_desurv_overlap.R` (Step 9), all under `results/benchmark_sim/`.

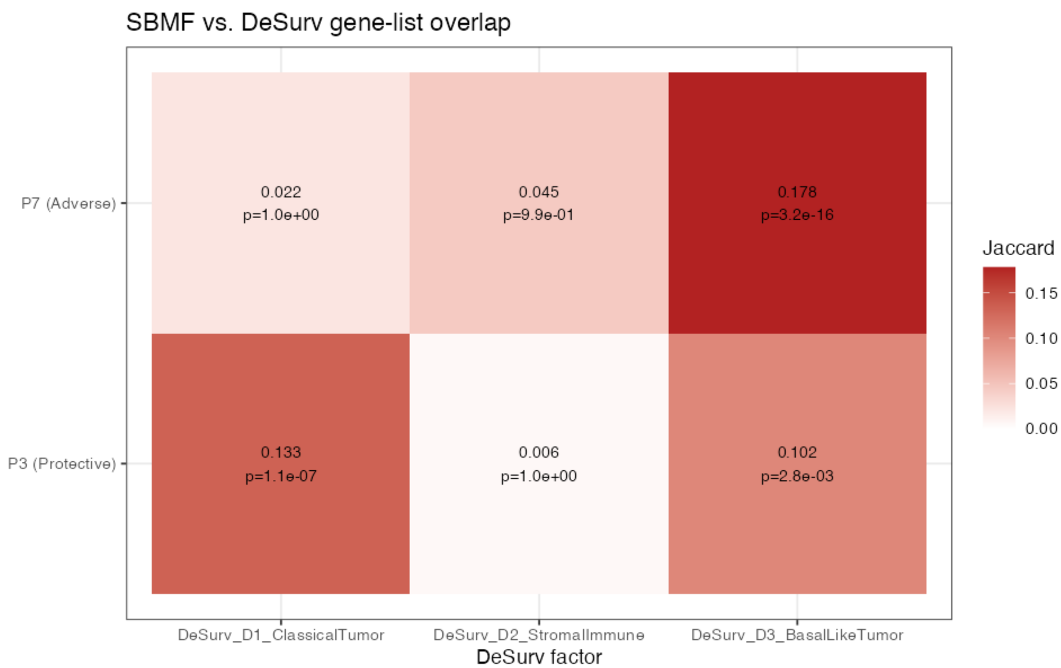

Figure 5: Jaccard overlap between each program's top-270 weighted genes and each DeSurv factor's gene list.